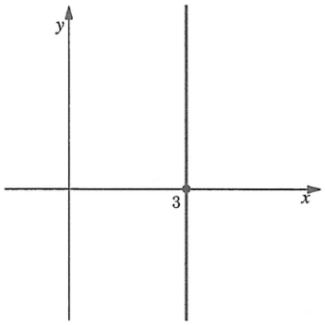
Graph Sketching: Linear Graphs

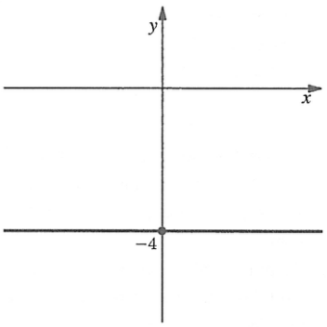
Questions 1–11

Question 1. Find the equation of each of the linear graphs below:

a)

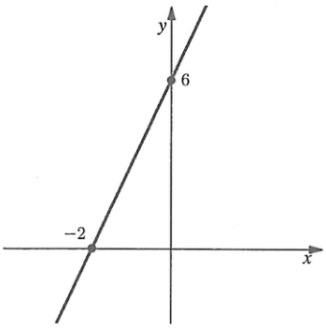


b)

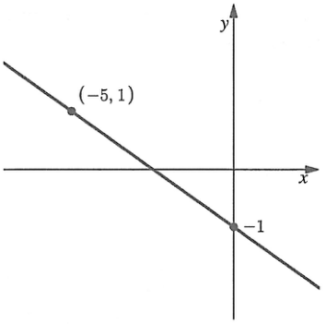


Question 2. Find the equation of each of the following graphs in the form $ax + by = c$, where a, b and c are integers.

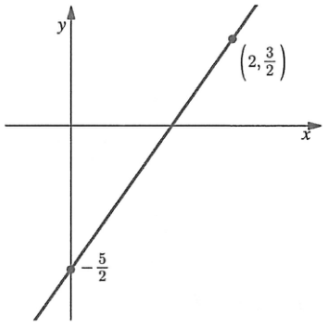
a)



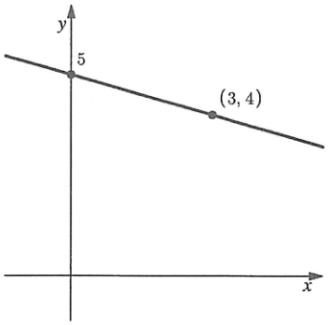
b)



c)



d)



Question 3. Sketch the following linear graphs, identifying where they intersect the coordinate axes:

a) $y = 2x - 7$

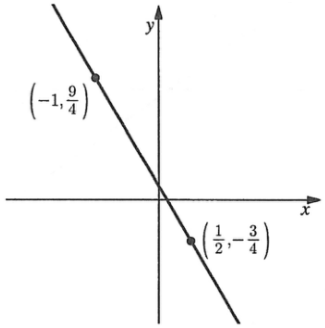
b) $y = 6 - 10x$

c) $y = \frac{1}{2}x + 5$

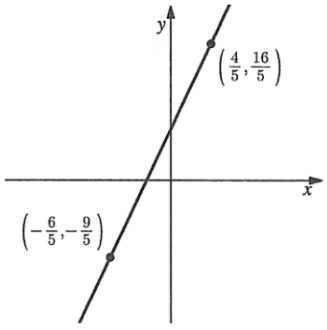
d) $y = 2 - \frac{1}{3}x$

Question 4. Find the equation of each of the following graphs in the form $ax + by = c$, where a, b and c are integers.

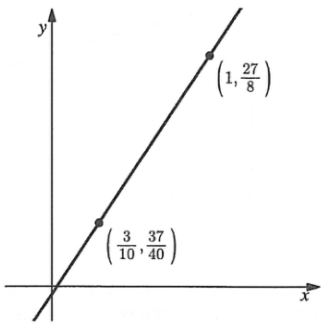
a)



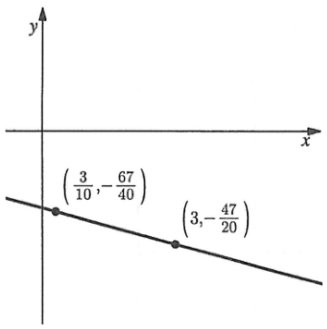
b)



c)



d)



Question 5. Sketch the following linear graphs, identifying where they intersect the coordinate axes:

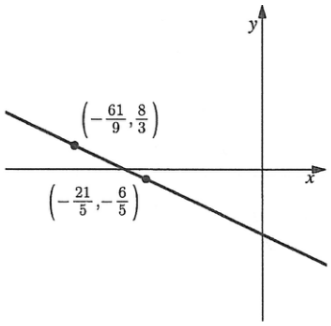
a) $2x + 5y = 10$
b) $4x - 9y = 36$
c) $-7x + 2y = 28$
d) $6x + 11y = -2$

Question 6.

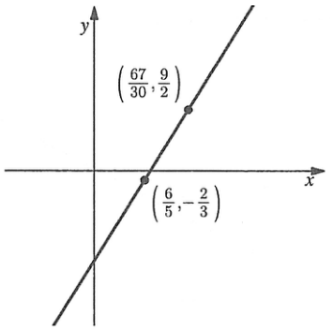
a) Sketch the graphs of $y = 2x + 8$ and $2y = 10 - x$ on the same axes, identifying where the graphs cross the coordinate axes.
b) Determine where the two lines intersect.
c) Find the area of the triangle formed by the two lines and the x-axis.
d) Find the area of the triangle formed by the two lines and the y-axis.

Question 7. Find the equation of each of the following graphs in the form $x + ay = b$, where a and b are constants.

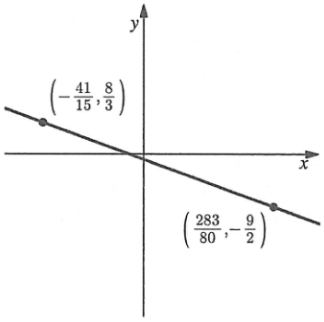
a)



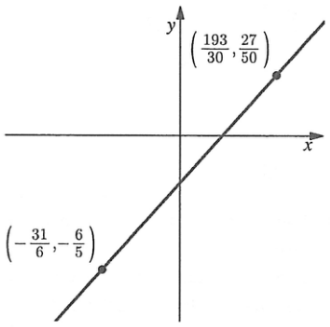
b)



c)



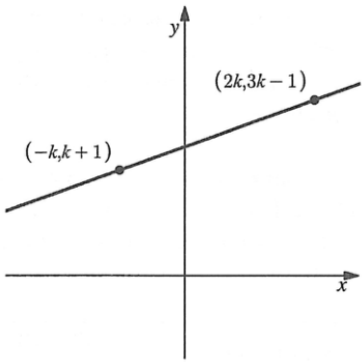
d)



Question 8. Sketch the following linear graphs, identifying where they intersect the coordinate axes:

a) $2y = 3x - 7$
b) $5y + 1 = 6x$
c) $3y - 3 = 8x$
d) $10y = 4 - 7x$

Question 9. The diagram below shows a linear function. Find the equation of the line in the form $y = ax + b$, where a and b are given in terms of k, where $k > 1$.



Question 10. Sketch the following linear graphs, identifying where they intersect the coordinate axes. It is given that $k > 0$.

a) $y = kx + 2$
b) $y = 8 - 2kx$
c) $ky = 3 - x$
d) $2x + 7y = k$
e) $3x + ky = 9$
f) $6kx - y = -3$
g) $\frac{x}{k} + \frac{y}{2k} = 4$

Question 11. The diagram below shows two lines, l_1 and l_2.

a) Find the equation of the line l_1.

b) Find the equation of the line l_2.

c) Determine the coordinates, in terms of k, of where l_1 and l_2 intersect.

